

RESEARCH ARTICLE

Optimal filters for the input conveyed by controllers using iterative learning strategies with alternating data losses and fading

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The time-varying nature of wireless channels leads to alternating data losses and fading, which pose significant challenges for ensuring the tracking accuracy of remotely controlled objects using iterative learning strategies. To address this challenge, we developed a filter to estimate the input conveyed by controllers through the time-varying channel. Firstly, a transmission model is formulated to depict the combined effects of alternating data losses and fading. Secondly, the formulated model is integrated with the given control strategy to form a filtering model. Thirdly, the filter based on projection theory is proposed. Finally, numerical results validate that feeding objects with the filtered input can significantly improve the tracking accuracy.

KEYWORDS

Iterative learning control, data loss, fading, time-varying channel, input filter

1 | INTRODUCTION

1.1 | Backgrounds

Iterative learning control (ILC) is a powerful strategy for controlling objects that move repetitively to track a specific trajectory. By utilizing previous tracking errors and appropriate learning gains to update input, this strategy significantly improves the object convergence in the iteration domain [1]. Moreover, unlike other control strategies, this strategy does not rely on precise mathematical models and can handle highly uncertain nonlinear dynamic systems. Since Arimoto first proposed this strategy, it has been extensively researched, particularly in the areas of learning algorithms [2], robustness [3, 4], and initial value problems [5].

Networked control systems (NCSs) have gained popularity alongside the advancement of information technology. In NCSs, controllers, actuators, and sensors are connected through a network, enabling real-time feedback in closed loops

[6]. NCSs provide several advantages over traditional control systems, including reduced system complexity and the resolution of issues related to equipment installation, maintenance, and high installation costs [7]. As a result, NCSs have found widespread applications in various domains, including intelligent transportation [8, 9], smart grid [10], aerospace [11], and unmanned driving [12].

However, the time-varying nature of wireless channels poses challenges in ensuring the tracking accuracy of objects controlled remotely using ILC strategies [13, 14]. Specifically, additive noise is inevitably present owing to the motion of free electrons. Furthermore, due to power attenuation, shadow occlusion, and multi-path effects, the channel gain is faded randomly between 0 and 1. Consequently, when the fading is extremely low, the data loss occurs as a specific manifestation. To overcome these challenges, new methods need to be developed to enhance the tracking accuracy of remotely controlled objects employing ILC strategies.

1.2 | Related works

Bernoulli variables and Markov chains are primarily employed to characterize the randomness of data losses. The former is commonly used to express the probability of a successfully received data packet. For instance, in [15], researchers used Bernoulli variables to describe the data loss problems of networked ILC (NILC) systems and transformed the design of ILC laws into the stability analysis of two-dimensional systems described by the Rosser model. In [16], the authors employed Bernoulli variables to design two compensation algorithms for nonlinear systems with output channel data losses. In [17], two independent Bernoulli variables were introduced to propose a compensation-based data-driven NILC method. In [18], for NCSs with data losses, a quantized ILC algorithm based on gradient was designed by using Bernoulli variables. Unlike this variable, the Markov chain can depict situations where the probability of data losses in network transmission changes over time. In [19], the data losses on both sides were represented as a Markov chain to analyze the convergence of NILC systems. In [20], a Markov chain was employed to describe the data loss scenario for linear time-varying systems. In [21], the authors applied a Markov chain to convert the NILC system with data losses into a switching system.

In addition, designing new learning strategies is currently the main solution aimed at ensuring the tracking accuracy of NILC systems with data losses. For example, in [22], a novel NILC scheme with a tuning factor was proposed to address the data losses in both input and output channels. In [23], a novel adaptive fault-tolerant NILC scheme was designed to address data losses on both input and output sides. In [24], the authors proposed a new NILC scheme for linear time-varying systems affected by data losses. In [25], researchers proposed a continuous updating scheme to address the issue of output data losses in NILC systems. In [26], a new NILC strategy with the direct feedback term was designed to address the random data loss issues. In addition to designing new learning strategies, the development of compensation schemes is another common method. It compensates for the loss of current data by utilizing the data previously obtained by systems. For instance, in [27, 28], the authors used data from the corresponding moment in the previous batch to compensate for losses. To address data losses, researchers predicted current data based on historical data for timely compensation [29, 30, 31]. In [32, 33], the data from previous moment in the same batch was used to substitute the lost data.

Fading also poses a challenge to ensure the tracking accuracy of NILC systems. For example, in [34], the authors used a diagonal matrix to represent multi-channel fading. In [35], a Kalman filtering method was proposed to derive a variable learning gain for addressing the impacts of fading on NILC

systems. In [36], researchers constructed a new variable step size ILC algorithm to address the fading problems. In [37], the authors developed a variable local average operator to guarantee the convergence of the NILC system with fading. Different from these works, the authors addressed the effects of fading from the perspective of input filtering [38]. Actually, a large number of optimal state filters were designed for NCSs under the effects of fading. For instance, in [39], a robust nonlinear filter was proposed to tackle the impacts of fading. In [40], a sliding average operator and a decreasing gain were introduced to correct the learning control algorithm for reducing the fading effect.

For NILC systems with additive noise, researchers have proposed many solutions. Additive noise includes not only channel noise from external transmission but also internal noise arising from state updating and output measurement. For example, in [41], two different filters were employed to address the impacts of additive noise on NILC systems. In [42], the authors designed a learning gain adjustment algorithm to improve the tracking accuracy and convergence rate of systems controlled by using iterative learning strategy with measurement noise. In [43], an optimal recursive filter was proposed to address the tracking problem of NILC systems with additive noise. In [44], a fast ILC method was developed to ensure the stability of switched discrete-time systems under the influence of additive noise. In [45], a Kalman filter-based learning algorithm was proposed to enhance the tracking accuracy of systems with stochastic noise. In [46], the authors analyzed the effects of additive noise and proposed a method to suppress these effects on NILC systems.

Here are several drawbacks identified in previous studies. First, to our knowledge, no study has simultaneously addressed the alternating effects of data losses and fading in time-varying channels. Secondly, in the aforementioned works, the primary methods for addressing communication interference involve designing new ILC strategies. Unfortunately, these methods would increase the control complexity. In addition to these drawbacks, which motivate this research, it should be noted that while various communication technologies have been proposed to ensure more reliable data transmission, these technologies are designed for wired or wireless communication systems, not for NILC systems. This means that residual errors are likely to accumulate as the iteration number increases, potentially impacting object tracking accuracy. Therefore, another motivating factor lies in seeking signal processing methods that align with the characteristics of NILC systems.

1.3 | Our contributions

In this paper, we investigate the tracking accuracy of objects remotely controlled employing ILC strategies. The differences

between this paper and existing works lie in two aspects. Firstly, it focuses on the combined influence of alternating data losses and fading in time-varying channels. Secondly, the main solution is to develop a filter for objects to estimate the input conveyed by controllers. When the controller is suitable for objects with ideal communication, the proposed filter improves the tracking accuracy of objects controlled under the effects of alternating data losses and fading. In summary, the main innovation of this paper lies in providing a novel approach to address the impact of communication constraints on the tracking accuracy of NILC systems. The advantage of this approach is that it can simultaneously handle multiple sources of interference across both links, while also avoiding the issue of poor adaptability caused by controller designs tailored to specific objects. To design the filter for estimating input conveyed by ILC controllers under the effects of alternating data losses and fading, the main works of this paper are reflected in the following aspects:

- Building a transmission model to capture the combined effects of alternating data losses and fading in time-varying channels;
- Developing a filtering model by integrating controller learning strategies with the data transmission model;
- Proposing a filter based on projection theory for objects to estimate the input conveyed by controllers.

The remaining sections are as follows: Section 2 outlines the initial problem, while Sections 3 and 4 focus on the design and verification of the filter. Finally, Section 5 summarizes this article. For the convenience of the following analysis, this article summarizes the nomenclature of variables in Table 1.

TABLE 1 The nomenclature of variables

A^T	the transpose of matrix A	A^{-1}	the inverse of matrix A
A^{-T}	the combination of inverse and transpose of matrix A	$Cov\{\cdot\}$	covariance
$(\cdot)$	the same contents as that in the previous round bracket	$E\{\cdot\}$	expectation
I	the identity matrix with appropriate dimensions	$Prob\{\cdot\}$	probability
$\hat{\cdot}$	estimated	$\tilde{\cdot}$	estimated error
$\bar{\cdot}$	mean		

2 | PROBLEM FORMULATION

Consider a controller as follows:

$$u_{k+1}^t(t) = u_k^t(t) + \Gamma(t)e_k^r(t+1) \quad (1)$$

where $e_k^r(t+1)$ represents the tracking error, $u_{k+1}^t(t)$ denotes the input conveyed to actuators, $\Gamma(t)$ is the gain that ensure the object convergence under ideal communication scenarios, k means the batch number, $t \in [0, 1, \dots, T-1]$ indicates the time in each batch.

We denote the tracking error conveyed by objects as $e_k^t(t)$ and the input data arriving at objects as $u_k^t(t)$. The transmission model for time-varying channels is described as follows:

$$u_k^r(t) = \xi_k(t)H_k(t)u_k^t(t) + (1 - \xi_k(t))u_{k-1}^r(t) + m_k(t) \quad (2)$$

$$e_k^r(t) = \eta_k(t)G_k(t)e_k^t(t) + (1 - \eta_k(t))e_{k-1}^r(t) + n_k(t) \quad (3)$$

where $H_k(t)$ and $G_k(t)$ are fading parameters; $\xi_k(t)$ and $\eta_k(t)$ are Bernoulli distributed variables; $1 - \xi_k(t)$ and $1 - \eta_k(t)$ mean the corresponding data is used as a substitute in the iteration domain when the channel gain is 0; $m_k(t)$ and $n_k(t)$ represent the presence of additive noise.

For the sake of exploration, we make some assumptions as follows:

Assumption 1. $m_k(t)$, $n_k(t)$, $H_k(t)$ and $G_k(t)$ are Gaussian distributed with known means and the variances of Q_m , Q_n , Q_H , and Q_G .

Assumption 2. $\eta_k(t)$ and $\xi_k(t)$ are Bernoulli distributed with probabilities $Prob\{\xi_k(t) = 1\} = \alpha$, $0 < \alpha \leq 1$, and $Prob\{\eta_k(t) = 1\} = \beta$, $0 < \beta \leq 1$, where α, β are known. Their variances are respectively expressed as Q_η and Q_ξ .

Assumption 3. $H_k(t)$, $G_k(t)$, $\eta_k(t)$, $\xi_k(t)$, $m_k(t)$ and $n_k(t)$ are uncorrelated.

Remark 1. Due to the random movement of massive free electrons in physical devices, additive noise always exists, and its statistical properties manifest as a Gaussian process. Furthermore, transmission gain varies between 0 and 1 due to the time-varying effects of wireless channels. When the time-varying effects are shallow, general fading occurs; when the time-varying effects are deep, the transmitted data cannot be demodulated. As a result, data loss occurs as a specific manifestation of fading. Therefore, these two communication uncertainties are alternating within wireless channels. The threshold for categorizing transmission gain attenuation as deep or shallow can be set based on practical circumstances. Without loss of generality, this paper assumes data loss occurs when the transmission gain is 0. Correspondingly, the general fading parameter ranges between $(0, 1]$.

It is clear that the input $u_k^r(t)$ arriving at actuators is disturbed by alternating data losses and fading in time-varying channels and cannot be used to guarantee the tracking accuracy. To this end, a filter would be designed for objects to estimate the input $u_k^t(t)$ conveyed by controllers. As long as the estimation is accurate, the filtered input $\hat{u}_{k|k}(t)$ can be applied to improve the tracking accuracy. To illustrate this idea clearly, it is shown in Figure 1.

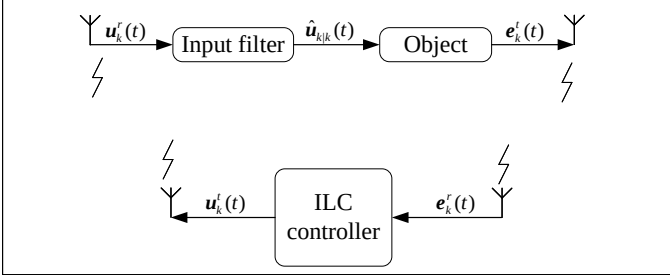


FIGURE 1 Diagram of objects remotely controlled by ILC strategies with input filters

3 | MAIN RESULTS

3.1 | Development of filtering models

To propose filters for the controlled objects, this subsection develops a filtering model. To be specific, the state updating process is described by combining the known controller learning strategies with the established transmission model to embrace all communication uncertainties, the output measurement process is represented by using the input transmission equation. Therefore, omitting the time index, the model is constructed as follows:

$$X_{k+1} = A_k X_k + B_{1,k} U_k + B_{2,k} W_k \quad (4)$$

$$Y_k = C_k X_k + V_k \quad (5)$$

where

$$\begin{aligned} X_{k+1} &= \begin{bmatrix} u_{k+1}^r(t) \\ u_k^r(t) \\ e_k^r(t+1) \end{bmatrix}, W_k = \begin{bmatrix} m_k(t) \\ n_k(t+1) \end{bmatrix}, U_k = e_k^r(t+1), \\ Y_k &= u_k^r(t), V_k = m_k(t), \\ A_k &= \begin{bmatrix} I & 0 & (1-\eta_k(t+1))\Gamma(t) \\ \xi_k(t)H_k(t)I & (1-\xi_k(t))I & 0 \\ 0 & 0 & (1-\eta_k(t+1))I \end{bmatrix}, \\ B_{1,k} &= \begin{bmatrix} \eta_k(t+1)G_k(t+1)\Gamma(t) \\ 0 \\ \eta_k(t+1)G_k(t+1)I \end{bmatrix}, \\ B_{2,k} &= \begin{bmatrix} 0 & \Gamma(t) \\ I & 0 \\ 0 & I \end{bmatrix}, C_k = [\xi_k(t)H_k(t)I \ (1-\xi_k(t))I \ 0]. \end{aligned} \quad (6)$$

Remark 2. According to the assumption about additive noise mentioned earlier, it is clear that $E\{W_k W_k^T\} = Q_W = \begin{bmatrix} Q_m & 0 \\ 0 & Q_n \end{bmatrix}$, $E\{V_k V_k^T\} = Q_V = Q_m$, $E\{W_k V_k^T\} = Q_S = \begin{bmatrix} Q_m \\ 0 \end{bmatrix}$. Additionally, according to the assumption about $\xi_k(t)$, $\eta_k(t)$, $H_k(t)$ and $G_k(t)$, it is clear that $E\{\xi_k(t)\} = \alpha$, $E\{\eta_k(t)\} = \beta$,

$E\{H_k(t)\} = \rho$, $E\{G_k(t)\} = \zeta$, $E\{\xi_k(t)\xi_k^T(t)\} = Q_\xi = \alpha(1-\alpha)$, and $E\{\eta_k(t)\eta_k^T(t)\} = Q_\eta = \beta(1-\beta)$.

3.2 | Filter design

To design the filter based on the constructed model (4) and (5), some lemmas are provided firstly.

Lemma 1. The following results that can be derived from the model (4) and (5):

$$\begin{aligned} \bar{A}_k &\triangleq E\{A_k\} = \begin{bmatrix} I & 0 & (1-\beta)\Gamma(t) \\ \alpha\rho I & (1-\alpha)I & 0 \\ 0 & 0 & (1-\beta)I \end{bmatrix}, \\ \bar{B}_{1,k} &\triangleq E\{B_{1,k}\} = \begin{bmatrix} \beta\zeta\Gamma(t) \\ 0 \\ \beta\zeta I \end{bmatrix}, \\ \bar{B}_{2,k} &\triangleq E\{B_{2,k}\} = \begin{bmatrix} 0 & \Gamma(t) \\ I & 0 \\ 0 & I \end{bmatrix}, \\ \bar{C}_k &\triangleq E\{C_k\} = [\alpha\rho I \ (1-\alpha)I \ 0], \end{aligned} \quad (7)$$

$$\begin{aligned} \Delta A_k &\triangleq A_k - \bar{A}_k = (\xi_k H_k - \alpha\rho)\Psi_1 + (\xi_k - \alpha)\Psi_2 + (\eta_k - \beta)\Psi_3, \\ \Delta B_{1,k} &\triangleq B_{1,k} - \bar{B}_{1,k} = (\eta_k G_k - \beta\zeta)\Psi_4, \\ \Delta B_{2,k} &\triangleq B_{2,k} - \bar{B}_{2,k} = 0, \\ \Delta C_k &\triangleq C_k - \bar{C}_k = (\xi_k H_k - \alpha\rho)\Psi_5 + (\xi_k - \alpha)\Psi_6 \end{aligned} \quad (8)$$

with

$$\begin{aligned} \Psi_1 &= \begin{bmatrix} 0 & 0 & 0 \\ I & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \Psi_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -I & 0 \\ 0 & 0 & 0 \end{bmatrix}, \Psi_3 = \begin{bmatrix} 0 & 0 & -\Gamma(t) \\ 0 & 0 & 0 \\ 0 & 0 & -I \end{bmatrix}, \\ \Psi_4 &= \begin{bmatrix} \Gamma(t) \\ 0 \\ I \end{bmatrix}, \Psi_5 = [I \ 0 \ 0], \Psi_6 = [0 \ -I \ 0]. \end{aligned}$$

Proof. According to the statistics of random variables given in Remark 2, equation (7) representing the mean of parameter matrices can be derived by taking an expectation on both sides of equation (6). As a result, equation (8) indicating the difference of parameter matrices can be obtained according to (6) and (7). $\square$

Lemma 2. The second-order moment of state $q_k = E\{X_k X_k^T\}$ meets the following recursion according to model (4) and (5):

$$\begin{aligned} q_{k+1} &= \bar{A}_k q_k \bar{A}_k^T + \bar{A}_k E\{X_k U_k^T\} \bar{B}_{1,k}^T + \bar{B}_{1,k} E\{U_k X_k^T\} \bar{A}_k^T \\ &\quad + \bar{B}_{1,k} \bar{U}_k \bar{U}_k^T \bar{B}_{1,k}^T + (Q_\xi Q_H + Q_\xi \rho^2 + Q_H \alpha^2) \Psi_1 q_k \Psi_1^T \\ &\quad + \alpha(1-\alpha) \Psi_2 q_k \Psi_2^T + \beta(1-\beta) \Psi_3 q_k \Psi_3^T \\ &\quad + (Q_\eta Q_G + Q_\eta \zeta^2 + \beta^2 Q_G) \Psi_4 \bar{U}_k \bar{U}_k^T \Psi_4^T + \bar{B}_{2,k} Q_W \bar{B}_{2,k}^T \end{aligned} \quad (9)$$

Proof. Consider the expectations on either side of (4), we obtain

$$X_{k+1} = (\bar{A}_k + \Delta A_k)X_k + (\bar{B}_{1,k} + \Delta B_{1,k})U_k + B_{2,k}W_k \quad (10)$$

From (10), it is clear that the input $u_k^t(t)$ affects the received output error $e_k^r(t+1)$, resulting in X_k and U_k being associated. The state matrix can be represented as

$$\begin{aligned} q_{k+1} = & \bar{A}_k q_k \bar{A}_k^T + \bar{A}_k E\{X_k U_k^T\} \bar{B}_{1,k}^T + \bar{B}_{1,k} E\{U_k X_k^T\} \bar{A}_k^T \\ & + \bar{B}_{1,k} \bar{U}_k \bar{U}_k^T \bar{B}_{1,k}^T + E\{\Delta A_k q_k \Delta A_k^T\} \\ & + E\{(\Delta B_{1,k} \bar{U}_k)(\bullet)^T\} + E\{B_{2,k} W_k W_k^T B_{2,k}^T\} \end{aligned} \quad (11)$$

where

$$\begin{aligned} E\{\Delta A_k q_k \Delta A_k^T\} = & (Q_\xi Q_H + Q_\xi \rho^2 + Q_H \alpha^2) \Psi_1 q_k \Psi_1^T \\ & + \alpha(1-\alpha) \Psi_2 q_k \Psi_2^T + \beta(1-\beta) \Psi_3 q_k \Psi_3^T \end{aligned} \quad (12)$$

$$\begin{aligned} E\{(\Delta B_{1,k} \bar{U}_k)(\bullet)^T\} = & (Q_\eta Q_G + Q_\eta \zeta^2 + \beta^2 Q_G) \Psi_4 \bar{U}_k \bar{U}_k^T \Psi_4^T \end{aligned} \quad (13)$$

$$E\{B_{2,k} W_k W_k^T B_{2,k}^T\} = \bar{B}_{2,k} Q_W \bar{B}_{2,k}^T \quad (14)$$

Substituting (12) ~ (14) into (11), we obtain (9). $\square$

With the two lemmas, the filter based on orthogonality theory can be proposed. We first denote the innovation in filtering as ε_k with covariance $\theta_k = E\{\varepsilon_k \varepsilon_k^T\}$, $K_k = E\{X_k \varepsilon_k^T\} \theta_k^{-1}$ is the gain for filtering, $L_k = E\{X_{k+1} \varepsilon_k^T\} \theta_k^{-1}$ is the gain for predicting, $P_{k|k}$ is the covariance matrix of filtering errors, and $P_{k+1|k}$ is the covariance matrix of one-step prediction errors.

Theorem 1. For models (4) and (5), the filter is proposed as

$$\hat{X}_{k|k} = \hat{X}_{k|k-1} + K_k \varepsilon_k \quad (15)$$

$$\hat{X}_{k+1|k} = \hat{X}_{k+1|k-1} + L_k \varepsilon_k \quad (16)$$

$$\varepsilon_k = Y_k - \hat{Y}_{k|k-1} \quad (17)$$

$$\hat{X}_{k+1|k-1} = \bar{A}_k \hat{X}_{k|k-1} + \bar{B}_{1,k} U_k \quad (18)$$

$$\hat{Y}_{k|k-1} = \bar{C}_k \hat{X}_{k|k-1} \quad (19)$$

$$K_k = P_{k|k-1} \bar{C}_k^T \theta_k^{-1} \quad (20)$$

$$L_k = (\bar{A}_k P_{k|k-1} \bar{C}_k^T + T_k) \theta_k^{-1} \quad (21)$$

$$\theta_k = \bar{C}_k P_{k|k-1} \bar{C}_k^T + J_k \quad (22)$$

$$\begin{aligned} P_{k+1|k} = & (\bar{A}_k - L_k \bar{C}_k) P_{k|k-1} (\bar{A}_k - L_k \bar{C}_k)^T + L_k Q_V L_k^T \\ & + \Omega_{1,k} + \Omega_{2,k} + \Omega_{3,k} - \Omega_{4,k} - \Omega_{4,k}^T \end{aligned} \quad (23)$$

$$P_{k|k} = P_{k|k-1} - K_k \theta_k K_k^T \quad (24)$$

where

$$\begin{aligned} T_k = & (Q_\xi Q_H + Q_\xi \rho^2 + Q_H \alpha^2) \Psi_1 q_k \Psi_5^T \\ & + \alpha(1-\alpha) \Psi_2 q_k \Psi_6^T + \bar{B}_{2,k} Q_S, \end{aligned}$$

$$\begin{aligned} J_k = & (Q_\xi Q_H + Q_\xi \rho^2 + Q_H \alpha^2) \Psi_5 q_k \Psi_5^T \\ & + \alpha(1-\alpha) \Psi_6 q_k \Psi_6^T + Q_V, \end{aligned}$$

$$\begin{aligned} \Omega_{1,k} = & (Q_\xi Q_H + Q_\xi \rho^2 + Q_H \alpha^2) (\Psi_1 - L_k \Psi_5) q_k (\Psi_1 - L_k \Psi_5)^T \\ & + \alpha(1-\alpha) (\Psi_2 - L_k \Psi_6) q_k (\Psi_2 - L_k \Psi_6)^T \\ & + \beta(1-\beta) \Psi_3 q_k \Psi_3^T, \end{aligned}$$

$$\Omega_{2,k} = (Q_\eta Q_G + Q_\eta \zeta^2 + \beta^2 Q_G) \Psi_4 \bar{U}_k \bar{U}_k^T \Psi_4^T,$$

$$\Omega_{3,k} = \bar{B}_{2,k} Q_W \bar{B}_{2,k}^T,$$

$$\Omega_{4,k} = \bar{B}_{2,k} Q_S L_k^T.$$

Proof. Based on the orthogonal projection theory, the filter should satisfy (15) ~ (17). Because $E\{W_k\} = 0$, we project the two sides of (4) onto the space generated by $(Y_1, Y_2, \dots, Y_{k-1})$ to have

$$\hat{X}_{k+1|k-1} = \bar{A}_k \hat{X}_{k|k-1} + \bar{B}_{1,k} U_k$$

Because $E\{V_k\} = 0$, we project the information on sides of (5) onto the subspace spanned by $(Y_1, Y_2, \dots, Y_{k-1})$ to obtain

$$\hat{Y}_{k|k-1} = \bar{C}_k \hat{X}_{k|k-1}$$

Defining $\tilde{X}_{k|k-1} = X_k - \hat{X}_{k|k-1}$ as the error in one-step prediction and substituting the model output in (5) and the output of one-step prediction (19) into (17), we have ε_k as

$$\varepsilon_k = \Delta C_k X_k + \bar{C}_k \tilde{X}_{k|k-1} + V_k \quad (25)$$

Observing that $\tilde{X}_{k|k-1} \perp V_k$, $X_k \perp V_k$, ΔC_k , $\tilde{X}_{k|k-1}$, and V_k are all zero mean values, the covariance matrix of innovation θ_k can be rewritten as

$$\begin{aligned} \theta_k = & E\{\varepsilon_k \varepsilon_k^T\} \\ = & E\{\Delta C_k q_k \Delta C_k^T\} + \bar{C}_k P_{k|k-1} \bar{C}_k^T + Q_V \\ = & (Q_\xi Q_H + Q_\xi \rho^2 + Q_H \alpha^2) \Psi_5 q_k \Psi_5^T + \alpha(1-\alpha) \Psi_6 q_k \Psi_6^T \\ & + \bar{C}_k P_{k|k-1} \bar{C}_k^T + Q_V \end{aligned}$$

So the equation (22) is proven right here.

By applying equation (25) and considering that $\tilde{X}_{k|k-1} \perp \hat{X}_{k|k-1}$ and $E\{V_k\} = 0$, the gain for filtering K_k is expressed as:

$$\begin{aligned} K_k = & E\{X_k \varepsilon_k^T\} \theta_k^{-1} \\ = & E\{X_k \tilde{X}_{k|k-1}^T\} \bar{C}_k^T \theta_k^{-1} \\ = & E\{\tilde{X}_{k|k-1} \tilde{X}_{k|k-1}^T\} \bar{C}_k^T \theta_k^{-1} \\ = & P_{k|k-1} \bar{C}_k^T \theta_k^{-1} \end{aligned}$$

Correspondingly, the gain for predicting L_k could be provided by

$$\begin{aligned} L_k &= E\{X_{k+1}\varepsilon_k^T\}\theta_k^{-1} \\ &= E\{A_kX_k\varepsilon_k^T + B_{1,k}U_k\varepsilon_k^T + B_{2,k}W_k\varepsilon_k^T\}\theta_k^{-1} \end{aligned} \quad (26)$$

Substituting (25) into $E\{A_kX_k\varepsilon_k^T\}$, we obtain

$$E\{A_kX_k\varepsilon_k^T\} = E\{A_kq_k\Delta C_k^T\} + E\{A_kX_k\tilde{X}_{klk-1}^T C_k^T\} \quad (27)$$

where

$$\begin{aligned} E\{A_kq_k\Delta C_k^T\} &= E\{\Delta A_kq_k\Delta C_k^T\} \\ &= E\{((\xi_kH_k - \alpha\rho)\Psi_1 + (\xi_k - \alpha)\Psi_2 \\ &\quad + (\eta_k - \beta)\Psi_3)q_k((\xi_kH_k - \alpha\rho)\Psi_5 + (\xi_k - \alpha)\Psi_6)^T\} \\ &= E\{(\xi_kH_k - \alpha\rho)\Psi_1q_k\Psi_5^T(\xi_kH_k - \alpha\rho)^T \\ &\quad + (\xi_k - \alpha)\Psi_2q_k\Psi_6^T(\xi_k - \alpha)^T\} \\ &= (Q_\xi Q_H + Q_\xi \rho^2 + Q_H \alpha^2)\Psi_1q_k\Psi_5^T + \alpha(1 - \alpha)\Psi_2q_k\Psi_6^T \\ E\{A_kX_k\tilde{X}_{klk-1}^T C_k^T\} &= \bar{A}_k E\{(\tilde{X}_{klk-1} + \hat{X}_{klk-1})\tilde{X}_{klk-1}^T\}\bar{C}_k^T \\ &= \bar{A}_k P_{klk-1} \bar{C}_k^T \end{aligned}$$

Because $\varepsilon_k(t)$ and $\eta_k(t)$ are not correlated, as are $\tilde{X}_{klk-1}$ and $\hat{X}_{klk-1}$, substituting (25) into $E\{B_{1,k}U_k\varepsilon_k^T\}$ and $E\{B_{2,k}W_k\varepsilon_k^T\}$ results in:

$$\begin{aligned} E\{B_{1,k}U_k\varepsilon_k^T\} &= E\{\Delta B_{1,k}U_k(\Delta C_kX_k + \bar{C}_k\tilde{X}_{klk-1} + V_k)^T\} \\ &= E\{\Delta B_{1,k}U_kX_k^T\Delta C_k^T\} \\ &= E\{(\eta_kG_k - \beta\zeta)\Psi_4U_kX_k^T((\xi_kH_k - \alpha\rho)\Psi_5 + (\xi_k - \alpha)\Psi_6)^T\} \\ &= 0 \end{aligned} \quad (28)$$

$$\begin{aligned} E\{B_{2,k}W_k\varepsilon_k^T\} &= E\{B_{2,k}W_k(\Delta C_kX_k + \bar{C}_k\tilde{X}_{klk-1} + V_k)^T\} \\ &= \bar{B}_{2,k}Q_S \end{aligned} \quad (29)$$

Putting (27) ~ (29) into (26), we obtain (21).

The error in one-step prediction $\tilde{X}_{k+1|k}$ is obtained by deducting $\hat{X}_{k+1|k}$ in (16) from X_{k+1} .

$$\begin{aligned} \tilde{X}_{k+1|k} &= (\bar{A}_k - L_k\bar{C}_k)\tilde{X}_{klk-1} + (\Delta A_k - L_k\Delta C_k)X_k \\ &\quad + \Delta B_{1,k}U_k + B_{2,k}W_k - L_kV_k \end{aligned}$$

Thus, the covariance matrix $P_{k+1|k}$ of prediction errors can be obtained as

$$\begin{aligned} P_{k+1|k} &= (\bar{A}_k - L_k\bar{C}_k)P_{klk-1}(\bar{A}_k - L_k\bar{C}_k)^T \\ &\quad + E\{(\Delta A_k - L_k\Delta C_k)q_k(\Delta A_k - L_k\Delta C_k)^T\} \\ &\quad + E\{\Delta B_{1,k}U_kU_k^T\Delta B_{1,k}^T\} + E\{B_{2,k}Q_WB_{2,k}^T\} \\ &\quad + L_kQ_VL_k^T + E\{(\Delta A_k - L_k\Delta C_k)X_kU_k^T\Delta B_{1,k}^T\} \\ &\quad + E\{(\Delta A_k - L_k\Delta C_k)X_kU_k^T\Delta B_{1,k}^T\}^T \\ &\quad - E\{B_{2,k}Q_SL_k^T\} - E\{B_{2,k}Q_SL_k^T\}^T \end{aligned} \quad (30)$$

According to the statistical characteristics of variables, we further have

$$\begin{aligned} E\{(\Delta A_k - L_k\Delta C_k)q_k(\Delta A_k - L_k\Delta C_k)^T\} &= (Q_\xi Q_H + Q_\xi \rho^2 + Q_H \alpha^2)(\Psi_1 - L_k\Psi_5) \\ &\quad \times q_k(\Psi_1 - L_k\Psi_5)^T \\ &\quad + \alpha(1 - \alpha)(\Psi_2 - L_k\Psi_6)q_k(\Psi_2 - L_k\Psi_6)^T \\ &\quad + \beta(1 - \beta)\Psi_3q_k\Psi_3^T \end{aligned} \quad (31)$$

$$\begin{aligned} E\{\Delta B_{1,k}U_kU_k^T\Delta B_{1,k}^T\} &= (Q_\eta Q_G + Q_\eta \zeta^2 + \beta^2 Q_G)\Psi_4\bar{U}_k\bar{U}_k^T\Psi_4^T \end{aligned} \quad (32)$$

$$E\{B_{2,k}Q_WB_{2,k}^T\} = \bar{B}_{2,k}Q_W\bar{B}_{2,k}^T \quad (33)$$

$$E\{(\Delta A_k - L_k\Delta C_k)X_kU_k^T\Delta B_{1,k}^T\} = 0 \quad (34)$$

$$E\{B_{2,k}Q_SL_k^T\} = \bar{B}_{2,k}Q_SL_k^T \quad (35)$$

Putting (31) ~ (35) into (30). From (15), the filtering error is given as

$$\tilde{X}_{klk} = \tilde{X}_{klk-1} - K_k\varepsilon_k$$

Therefore, the covariance matrix P_{klk} of filtering errors can be given by

$$P_{klk} = P_{klk-1} + K_k\theta_kK_k^T - K_kE\{\varepsilon_k\tilde{X}_{klk-1}^T\} - E\{\tilde{X}_{klk-1}\varepsilon_k^T\}K_k^T \quad (36)$$

In addition, according to the definition of K_k , we get

$$\begin{aligned} E\{\tilde{X}_{klk-1}\varepsilon_k^T\} &= E\{X_k\varepsilon_k^T\} \\ &= K_k\theta_k \end{aligned} \quad (37)$$

Substituting (37) into (36), (24) is proven. $\square$

Remark 3. The purpose of this paper is to design filtering methods for existing ILC controllers, which cannot be directly used to guarantee the tracking accuracy of objects under the effects of alternating data losses and fading in time-varying channels. Although the filtering method proposed in this paper is based on proportional learning strategies for controllers, according to the design idea represented in this paper, filters can

be also derived for other ILC controllers designed for initial value errors, non-repetitive trajectory tracking and so on.

4 | NUMERICAL SIMULATION

4.1 | A linear example

A linear object is given as:

$$\begin{cases} x_k(t+1) = \begin{bmatrix} -0.5 & 0 & 0 \\ 1 & 1.24 & -0.87 \\ 0 & 0.87 & 0 \end{bmatrix} x_k(t) + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u_k^r(t) \\ y_k^t(t) = \begin{bmatrix} 2 & 2.6 & -2.8 \end{bmatrix} x_k(t) \end{cases}$$

where $x_k(t)$ is the object state, $u_k^r(t)$ is the input arriving at objects, $y_k^t(t)$ is the output conveyed by objects.

The target trajectory is defined as

$$y_d(t) = 2.5 \times (1 + \sin(2\pi t/T) - \pi/2)$$

The controller provided in (1) is used with an initial state of $x_k(0) = [0 \ 0 \ 0]^T$ and an initial input of $u_0(t) = 0$. The operation period is set to $T = 100$, and the learning gain $\Gamma(t)$ is selected as 0.1, ensuring object convergence under ideal communication scenarios. The mean and variance of $m_k(t)$ and $n_k(t)$ are 0 and 0.02, respectively. The mean and variance of $H_k(t)$ and $G_k(t)$ are 0.9 and 0.01, respectively. The probabilities of data losses are given as $\alpha = \beta = 0.92$. To the proposed filter, the initial values are set as $\hat{X}_{0|1} = [1 \ 1 \ 1]^T$ and $P_{0|1} = 1I_3$, where I_3 is the identify matrix. Additionally, besides the raw input received on the side of objects, the inputs obtained using iteration-compensation or Kalman filtering [47, 48, 49] are utilized as two benchmarks in this simulation.

Figure 2 exhibits that the raw input received on the side of this linear object is significantly affected by alternating data losses and fading in time-varying channels. As a consequence, the corresponding output is unable to accurately trace the expected trajectory at the 25-th iteration, as shown in Figure 3. Figure 4 depicts the input received with data compensation. It can be seen that although the fluctuation of this input curve is partially suppressed, the output still does not meet the expected trajectory, as shown in Figure 5. The reason is that although the issue of data losses has been addressed by using data compensation, the transmission process is still influenced by additive noise and fading. Figure 6 displays the input estimated by the Kalman filter. It can be seen that although the fluctuation of this input curve is further partially suppressed, the output still does not meet the expected trajectory, as shown in Figure 7. The reason is that although the issue of additive noise has been addressed by using the Kalman filter, the transmission process

is still influenced by fading. Proceeding to Figure 8, it illustrates that the input produced by the proposed filter achieves complete convergence at the 25-th iteration despite the presence of these communication interference. The reason is that the proposed filter takes all communication interference in both channels into account. Therefore, Figure 9 shows that the output accurately converges to the expected trajectory.

Additionally, the absolute mean of input errors is used as a metric to highlight the benefits of the developed method. According to Figure 10, it is clear that only the absolute mean of input errors produced by the proposed filtering method converges. This is because typical data compensation methods in existing literature only address data losses, while the Kalman filtering method mainly addresses the problem of additive noise. In a similar manner, Figure 11 illustrates that only the absolute mean of the output errors produced by the proposed filtering method converges.

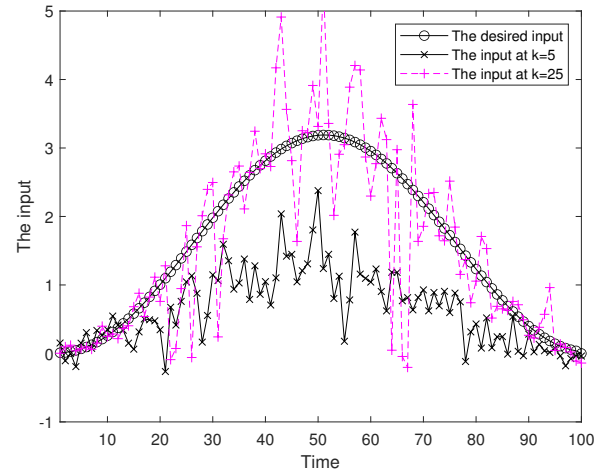


FIGURE 2 The raw input received on the side of objects

4.2 | A nonlinear example

A nonlinear object is given as:

$$\begin{cases} x_k(t+1) = 0.1 \sin(x_k(t)) + 0.5u_k(t) \\ y_k(t) = 4.5x_k(t) \end{cases}$$

The target trajectory is defined as

$$y_d(t) = 0.05 \times (2\pi t^2/T)$$

where initial condition is set as $x_{1,k}(t) = x_{2,k}(t) = 0$, $u_k(0) = 0$. $T = 100$. $\Gamma(t) = 0.1$, which ensures the object convergence under ideal communication scenarios. The mean and

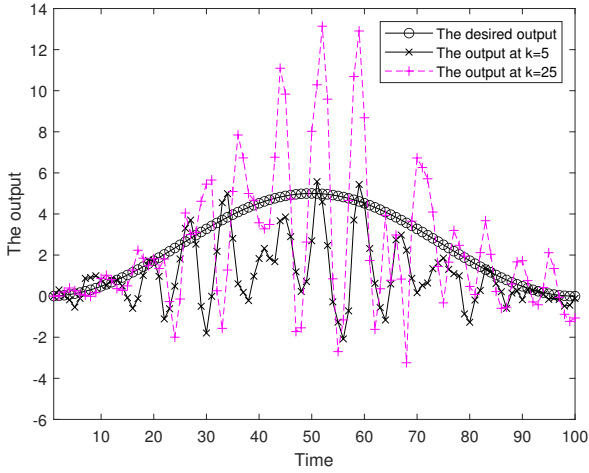


FIGURE 3 The output produced by raw input

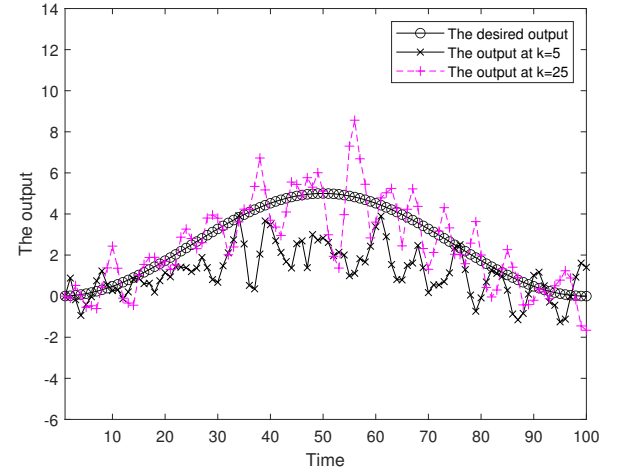


FIGURE 5 The output produced by the input received with data compensation

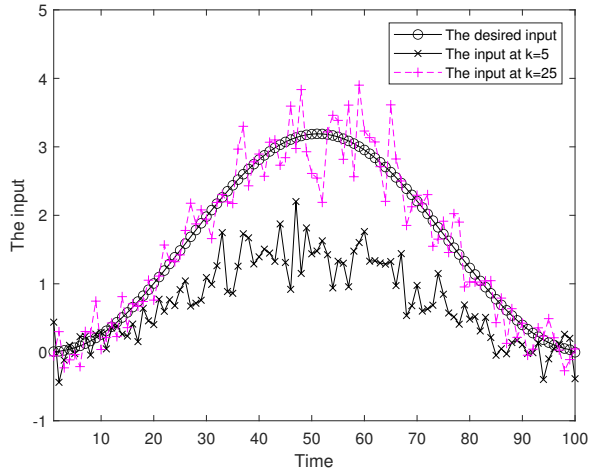


FIGURE 4 The input received with data compensation

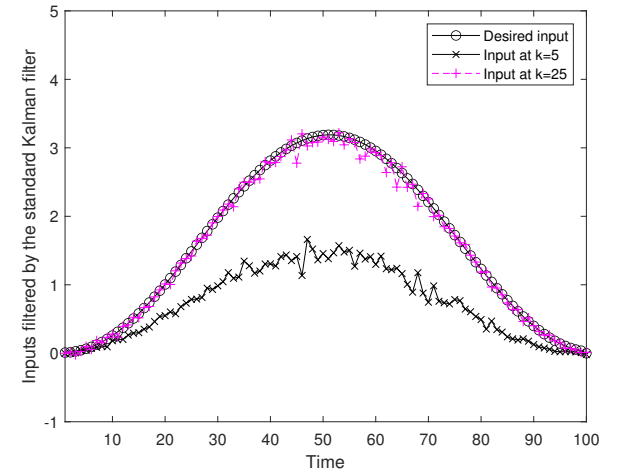


FIGURE 6 The input estimated by the Kalman filter

variance of $m_k(t)$ and $n_k(t)$ are 0 and 0.1, respectively. The mean and variance of the $H_k(t)$ and $G_k(t)$ are 0.9 and 0.01, respectively. The probabilities of data losses and the initial values of the proposed filter are set to the same as in the previous example. Additionally, the same benchmarks are also used in this example.

Figure 12 depicts how alternating data losses and fading in time-varying channels affect the raw input received on the side of this nonlinear object. Due to this input fluctuating strongly, the output of this nonlinear object cannot track the desired trajectory, as demonstrated in Figure 13. Figure 14 depicts the input received with data compensation. Despite the effective resolution of the data loss issues, the output of this nonlinear object still fails to track the desired trajectory, as shown in Figure 15. This indicates that even after addressing the

data loss issues, additive noise and fading continuously affect the input received with the iteration-compensation approach. Figure 16 displays the input estimated by the Kalman filter. Despite the effective resolution of the additive noise issues, the output of this nonlinear system still does not successfully follow the desired trajectory, as shown in Figure 17. This indicates that even after addressing the additive noise issues, data losses and fading continuously affect the input received with the iteration-compensation approach. Figure 18 displays the input produced by the proposed filtering method. Despite the various communication interference in both channels, the designed filter ensures the input estimated on the object side to

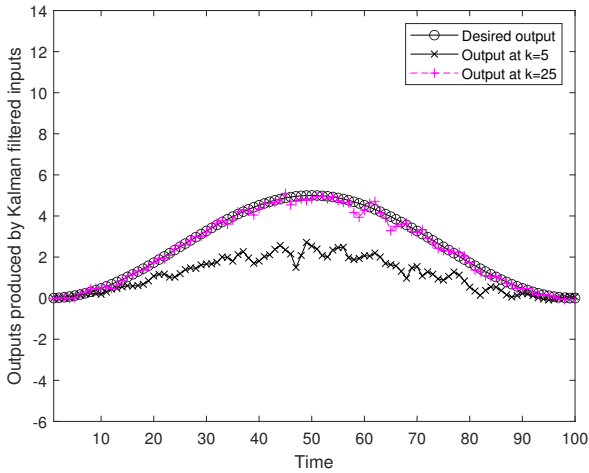


FIGURE 7 The output produced by the estimated input of the Kalman filter

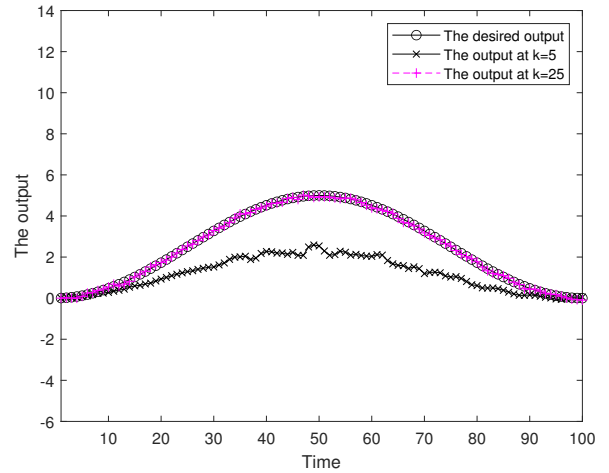


FIGURE 9 The output produced by the estimated input of the designed filter

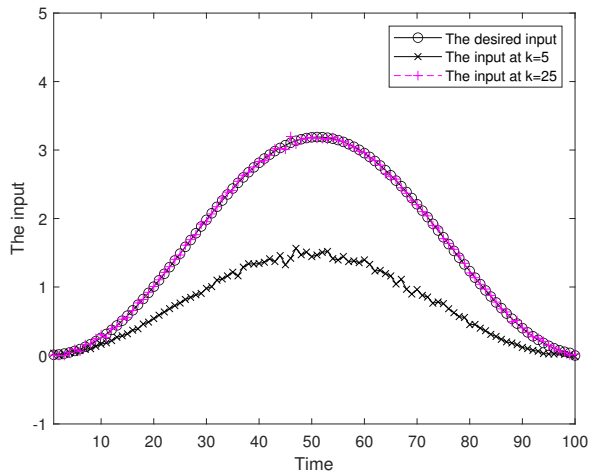


FIGURE 8 The input estimated by the designed filter

achieve complete convergence at the 25 – *th* iteration. Consequently, the output precisely traces the expected trajectory, as demonstrated in Figure 19.

Furthermore, the scenarios shown in Figures 20 and 21 are similar to those in Figures 10 and 11. As shown in Figure 20, only the absolute mean of input errors generated by the proposed method converges. Likewise, Figure 21 demonstrates that the absolute mean of output errors also converges as the iterations progress.

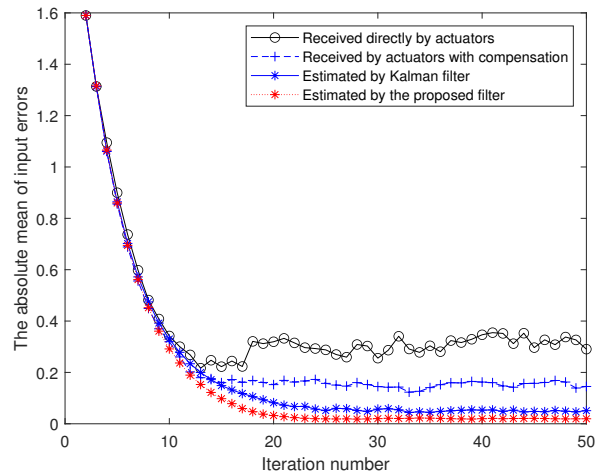


FIGURE 10 The comparison of the absolute mean of input errors

5 | CONCLUSION

To ensure the convergence of objects under conditions of alternating data losses and fading in time-varying channels, a filter has been developed using both controller learning strategies and data transmission models to estimate the input conveyed by the controller adopting ILC strategies. The efficiency of the proposed filter has been demonstrated in simulation results.

For objects controlled remotely, uncertainties arise not only during data transmission but also at the transmitters, such as quantization errors. Additionally, the filtering method proposed in this paper relies on the statistics of additive noise. In

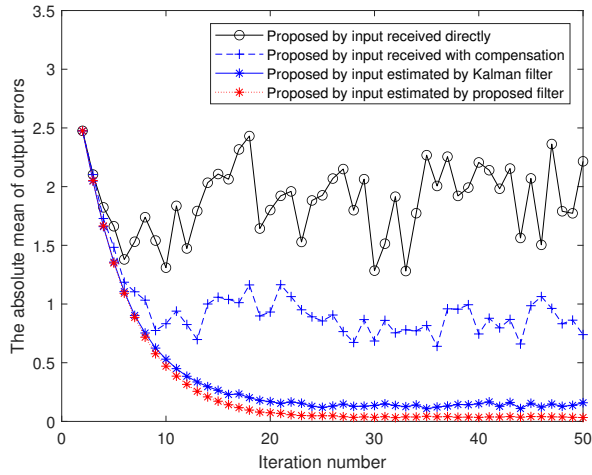


FIGURE 11 The comparison of the absolute mean of output errors

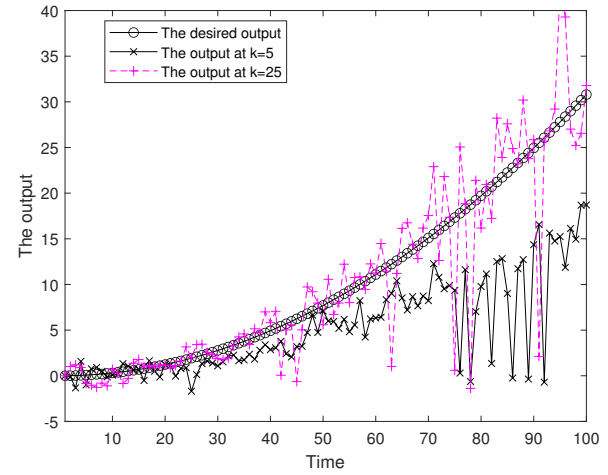


FIGURE 13 The output produced by the raw input

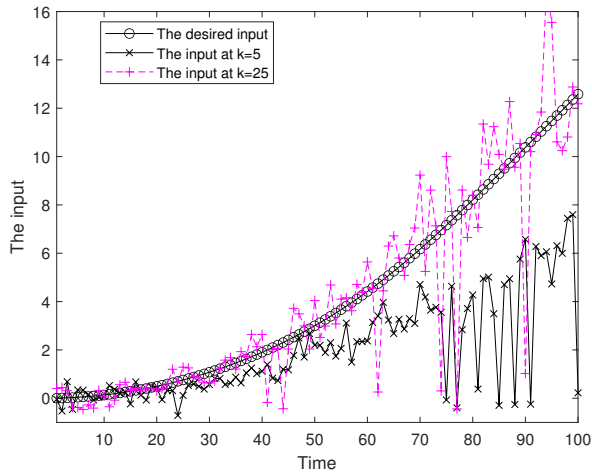


FIGURE 12 The raw input received on the side of objects

the future, we will further investigate the impact of the entire communication link and develop more robust filtering methods. In addition, we will explore better methods for calculating the filtering gain to enhance the convergence speed of designed filters.

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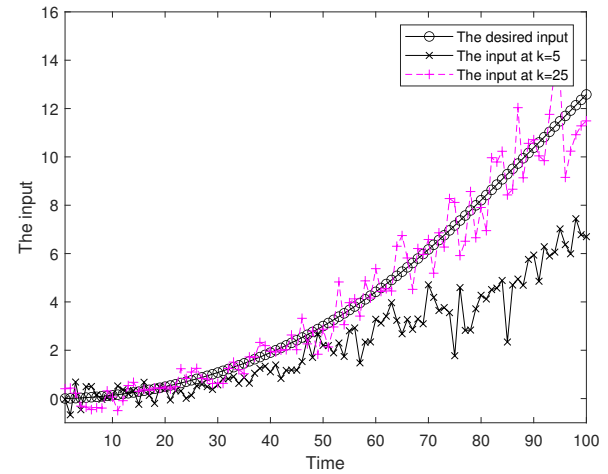


FIGURE 14 The input received with data compensation

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AUTHOR CONTRIBUTIONS

Lixun Huang: Conceptualization (lead); Methodology (supporting); Writing-original draft (equal); Writing-review and editing (equal). Huan Wang: Conceptualization (supporting); Methodology (lead); Writing-original draft (equal). Lijun Sun: Conceptualization (supporting); Writing-review and editing

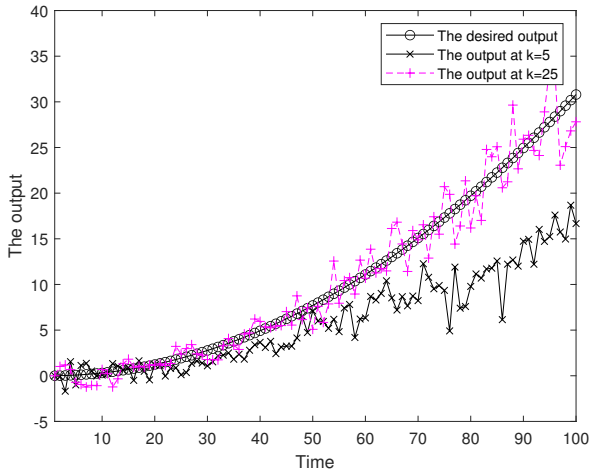


FIGURE 15 The output produced by the input received with data compensation

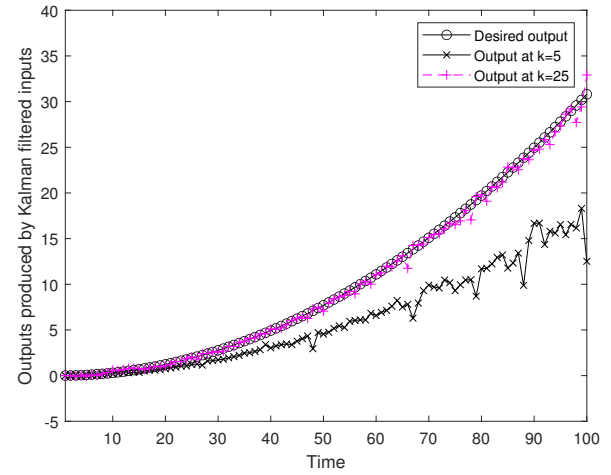


FIGURE 17 The output produced by the estimated input of the Kalman filter

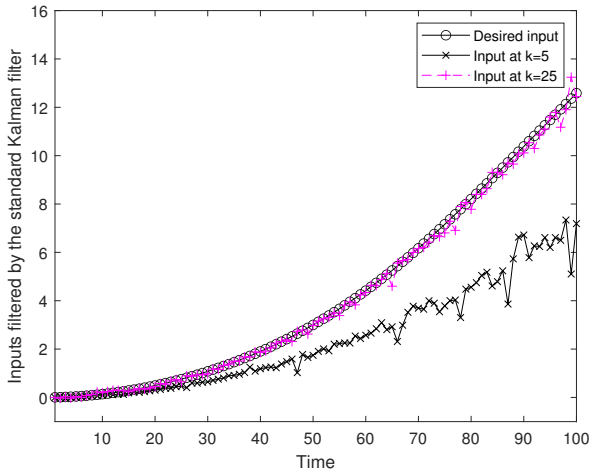


FIGURE 16 The input estimated by the Kalman filter

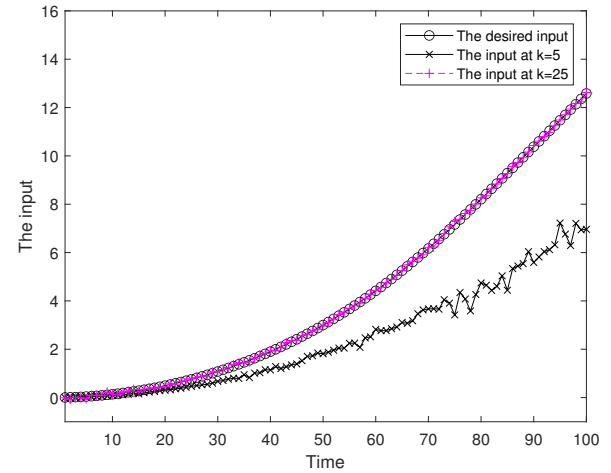


FIGURE 18 The input estimated by the designed filter

(equal). Tianfei Chen: Conceptualization (supporting). Qiwen Zhang: Methodology (supporting). Bin Jiang: Methodology (supporting).

CONFLICT OF INTEREST

The authors declare no potential conflict of interests.

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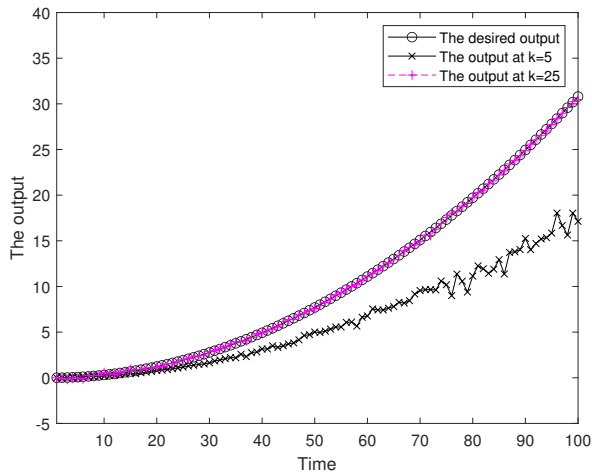


FIGURE 19 The output produced by the estimated input of the designed filter

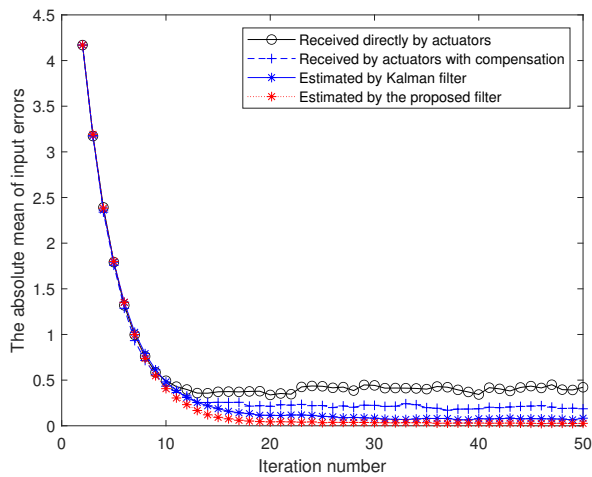


FIGURE 20 The comparison of the absolute mean of input errors

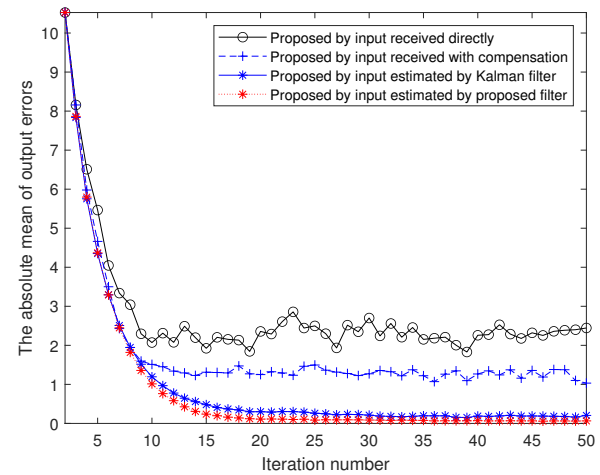


FIGURE 21 The comparison of the absolute mean of output errors

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